

Yanfei Zhu

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EDUCATION

University of Wisconsin-Madison

Madison, WI

M.A. in Mathematics, GPA: 3.844/4.000

Sep. 2024 – May 2026

Visiting International Student Program (VISP)

- **Courses:** Introductory Topology I&II (Algebraic Topology), Homological Algebra, Abstract Algebra, Modern Algebra
- Began UW coursework during final year of undergraduate studies; returned to SUSTech for B.S. defense; resumed UW to complete M.A.

Southern University of Science and Technology (SUSTech)

Shenzhen, CN

B.S. in Mathematics and Applied Mathematics

Sep. 2021 – Jun. 2025

- **Weighted Average Score:** 88.09/100
- **Courses:** Algebraic Topology, Topology, Representation of Groups, Abstract Algebra, Advanced Linear Algebra, Discrete Mathematics, Elementary Number Theory

Research Interests: Equivariant homotopy theory, Homological stability, Differential geometrics & topology

RESEARCH EXPERIENCES

Directed Reading on Intersection Homology

Aug. 2025 – Dec. 2025

Advisor: Prof. Laurentiu Maxim, Mathematics Department, UW-Madison

Topics: Intersection Homology, Sheaves on Topological Spaces, Verdier Duality

Directed Reading on Riemannian Geometry and Morse Theory

Mar. 2025 – Oct. 2025

Advisor: Prof. Ruobing Zhang, Mathematics Department, UC San Diego

Topics: Riemannian geometry (Petersen, do Carmo), Morse theory (Milnor)

Directed Reading on Homological Stability

Apr. 2025 – Jul. 2025

Advisor: Prof. Yifei Zhu, Mathematics Department, SUSTech

Topics: Homological Stability, Configuration Spaces, FI-modules, Braid Groups

Undergraduate Thesis: Symmetry of Solutions in Nonlinear Aggregation-Diffusion Equations and Its Applications

Dec. 2024 – Jul. 2025

Advisor: Prof. Linlin Su, SUSTech

- Studied the symmetry properties of stationary solutions via nonlinear aggregation-diffusion equations and focused on their radial symmetry and applications in numerical methods.
- The results verified the effectiveness of the use of symmetrization in numerical solutions, providing new insights for modeling collective behavior in mathematical biology and related fields.

Summer School at the University of Science and Technology of China, Hefei, CN

July 2025

- Topic in geometry and analysis

IWoAT Summer School at Westlake University, Hangzhou, CN

July 2025

- Topics: Computations with the Adams Spectral Sequence and the Last Kervaire Invariant Problem

Reading Seminar on Homological Algebra

Sep. 2023 – Dec. 2023

Advisor: Prof. Ziqing Xiang, Mathematics Department, SUSTech

Topics: Category Theory, Homological Algebra

TEACHING EXPERIENCE

Grader, Department of Mathematics, UW–Madison

Feb. 2026 – May. 2026

- Grading undergraduate mathematics assignments; providing clear written feedback.

ADDITIONAL SKILLS

- **Technical Skills:** LaTeX, Python, MATLAB, Java
- **Languages:** Mandarin (native), English (fluent), Japanese (basic)